Linux Setup Guide: Markdown to PDF with Math

Companion to `md_math_to_pdf.py` · Ubuntu/Debian commands verified against Ubuntu 24.04 LTS (pandoc 3.1.3, TeX Live 2023) · 8 October 2026

1. What you need

`md_math_to_pdf.py` cleans up the Markdown and then hands it to two external programs, pandoc and XeLaTeX. The script itself uses only Python's standard library (Python 3.8 or newer; tested with 3.12). Neither pandoc nor a TeX engine is installed by default on a standard Ubuntu install, so a fresh machine needs the steps below.

Component	Purpose	Ubuntu/Debian package
pandoc	Converts Markdown to LaTeX and drives the PDF build	`pandoc`
XeLaTeX	Typesets the PDF, including the equations	`texlive-xetex`
Core LaTeX packages	amsmath, fontspec, unicode-math, geometry, hyperref and similar	`texlive-latex-recommended`, `texlive-fonts-recommended`
Latin Modern font package	Loaded by pandoc's template; not pulled in by the packages above	`lmodern`
Extra LaTeX packages (optional)	Long URLs wrap correctly; code blocks keep straight quotes	`texlive-latex-extra`

2. Install on Ubuntu, Debian, Linux Mint or Pop!_OS

```
sudo apt update
sudo apt install -y pandoc texlive-xetex texlive-latex-recommended \
    texlive-fonts-recommended texlive-latex-extra lmodern
```

Disk space. About 0.5 GB including dependencies (pandoc is roughly 190 MB; the TeX packages roughly 260 MB). `texlive-latex-extra` is the largest TeX piece at about 80 MB; you can leave it out if space matters, and PDFs will still build. Sizes come from apt metadata on Ubuntu 24.04 and differ slightly between releases.

The `lmodern` gotcha. On Ubuntu 24.04, `texlive-fonts-recommended` alone does not provide `lmodern.sty`, and pandoc's LaTeX template always loads it. Without the separate `lmodern` package the build stops with a "file lmodern.sty not found" error, so keep `lmodern` in the install command.

About the pandoc version. Distribution repositories lag behind upstream. Ubuntu 24.04 ships pandoc 3.1.3, which works with the script. Older releases ship pandoc 2.x, which should also work but has not been tested here. For the latest release, download the `.deb` for your CPU (amd64 or arm64) from the pandoc releases page on GitHub and install it:

```
sudo apt install ./pandoc-<version>-1-amd64.deb
```

3. Verify the installation

```
pandoc --version | head -1
xelatex --version | head -1
python3 --version
```

Expected output looks like `pandoc 3.1.3`, `XeTeX 3.141592653-2.6-0.999995 (TeX Live 2023/Debian)` and `Python 3.12.3` (your versions may differ).

Then run a one-minute smoke test that does not involve the project script:

```
cat > smoke.md <<'EOF'
# Smoke test

Inline math: $E = mc^2$

$$\sum_{i=1}^{n} x_i^2$$
EOF
pandoc smoke.md -o smoke.pdf --pdf-engine=xelatex
```

If `smoke.pdf` appears with a heading, an inline equation and a display equation, the toolchain is complete. Finally, convert a real file:

```
python3 md_math_to_pdf.py chapter2.md          # writes chapter2.pdf
python3 md_math_to_pdf.py chapter2.md --toc -o notes.pdf
```

Use the script rather than plain pandoc for your notes. Plain pandoc does not understand the `\\(...\\)` and `\\[...\\]` delimiters, and it fails on `\boldsymbol` with XeLaTeX; the script fixes both (see Troubleshooting).

4. Other Linux distributions

These commands were not tested for this guide. Package names change between releases, so search for the exact names if one is not found (`dnf search texlive-xetex`, `pacman -Ss texlive`).

Distribution	Command
Fedora, RHEL, Rocky, Alma	`sudo dnf install pandoc texlive-xetex texlive-collection-latexrecommended texlive-collection-fontsrecommended`
Arch, Manjaro	`sudo pacman -S pandoc-cli texlive-xetex texlive-latexrecommended texlive-fontsrecommended texlive-latexextra`

5. Lighter TeX option: TinyTeX (any distribution, no root)

TinyTeX is a small TeX Live that downloads LaTeX packages on demand. This route was not tested for this guide, and pandoc must still be installed separately (section 2).

```
wget -qO- "https://yihui.org/tinytex/install-bin-unix.sh" | sh
```

Open a new terminal and run `xelatex --version`. If the command is not found, add the folder the installer reported to your `PATH`; if XeTeX itself is missing, run `tlmgr install xetex`. When a build stops because a file is missing, find the package that provides it and install it:

```
tlmgr search --global --file "/<name>.sty"
tlmgr install <package-name>
```

6. If you cannot install LaTeX

HTML route (pandoc only). This produces a web page with the equations as MathML. Open it in a browser and use Print, then Save as PDF.

```
python3 md_math_to_pdf.py chapter2.md --html
```

Pure-Python route (no pandoc, no TeX). Use `md_math_to_pdf_pure.py`, which needs `md_math_to_pdf.py` in the same folder. Ubuntu 23.04 and later (and Debian 12) protect the system Python, so install the libraries in a virtual environment:

```
sudo apt install -y python3-venv
python3 -m venv ~/mdpdf-env
source ~/mdpdf-env/bin/activate
pip install markdown-it-py matplotlib reportlab pillow
python md_math_to_pdf_pure.py chapter2.md
```

This renderer understands only a subset of LaTeX math. Equations it cannot parse, such as matrices, are printed as raw TeX in red and listed at the end of the run.

7. Troubleshooting

Symptom	Cause	Fix
`pandoc: command not found`	pandoc is not installed	`sudo apt install pandoc`
`xelatex not found. Please select a different --pdf-engine or install xelatex`	XeTeX is not installed	`sudo apt install texlive-xetex`
LaTeX error: file `lmodern.sty` not found	The `lmodern` package is missing; `texlive-fonts-recommended` does not provide it	`sudo apt install lmodern`
LaTeX error: file `<name>.sty` not found (any other name)	Another LaTeX package is missing	`sudo apt install texlive-latex-extra`. If it persists, run `sudo apt install apt-file`, then `sudo apt-file update`, then `apt-file search <name>.sty` and install the package it lists

Symptom	Cause	Fix
`Improper alphabetic constant` mentioning `\mitphi`	`\boldsymbol` clashes with XeLaTeX's unicode-math package	Use `md_math_to_pdf.py`, which patches this automatically. When calling pandoc directly, pass a header file with `-H header.tex` containing `\AtBeginDocument{\renewcommand{\boldsymbol}[1]{\symbfit{#1}}}`
`Missing character: There is no ... in font`	Emoji or symbols the LaTeX font lacks	The script removes emoji by default; check the source for other unusual symbols
Equations appear as literal `\\(...\\)` text	The file was converted with plain pandoc	Run it through `md_math_to_pdf.py`
`error: externally-managed-environment`	The system Python is protected (PEP 668)	Install packages inside a virtual environment (section 6)
`ModuleNotFoundError: No module named 'md_math_to_pdf'`	`md_math_to_pdf_pure.py` was run without `md_math_to_pdf.py` beside it	Keep both files in the same folder

8. Uninstall

```
sudo apt remove --purge pandoc texlive-xetex texlive-latex-recommended \
    texlive-fonts-recommended texlive-latex-extra lmodern
sudo apt autoremove --purge
```

9. Quick reference

```
# 1. Install (Ubuntu/Debian)
sudo apt update && sudo apt install -y pandoc texlive-xetex texlive-latex-
recommended \
    texlive-fonts-recommended texlive-latex-extra lmodern

# 2. Check
pandoc --version | head -1 && xelatex --version | head -1

# 3. Convert
python3 md_math_to_pdf.py chapter2.md               # chapter2.pdf
python3 md_math_to_pdf.py chapter2.md --toc -o notes.pdf
python3 md_math_to_pdf.py chapter2.md --emit-md     # inspect the cleaned
Markdown only
```